



WFA Forensic Entomology Photography Standard Operating Procedure (SOP)

This SOP was designed to ease dataset additions for the forensically relevant dipterans classifiers developed by the WFA.

Equipment

- Microscope (preferably with a built-in camera/image function)
- 0.3-0.5 mm entomology pins (Grade 00-3)
- Soft-touch tweezers
- (Black/white) plastazote foam for the insertion of the pins
- Specimens stored with proper entomological care

Preparation

Use a specimen size-appropriate pin. Carefully mount the fly by sticking the pin through the second segment of the thorax and ensure the pin is inserted slightly off center. Use a pair of soft-touch tweezers to manipulate the specimen where necessary. Use a pinning block to place specimen at appropriate position on the pin. Stick the pin into appropriate plastazote foam. Use soft-touch tweezers or rotate the pin head with fingers to manipulate the specimen for different shots.

Photography

Each fly should be photographed from 6 different angles, focusing on: the head (anterior - face), the dorsal side, left and right lateral aspects, the calypters, and the wing (see reference photographs). Ensure the fly is oriented in parallel to the microscope, as it helps to achieve the best focus. Adjust the lighting and focus where necessary to get a clear, sharp picture of the target area. Be mindful of exposure levels, as overexposure can result in saturation or loss of color, whereas underexposure can make it difficult to recognize important details. Examples of overexposed, underexposed, and correctly lit images are shown below.

Storage

Overlapping file names can cause significant issues in the machine learning pipeline, hence, it is important to stick to the following format: yyyy-mm-dd_sample_n.n_area, all in lower case. Example: 2026-01-23_sample_3.4_head. For the area choose between head, dorsal, lateral, calypters, and wing. To ensure that the species do not get mixed

up it is important to save all the photos of samples belonging to the same species in their separate folders. The following folders already exist:

Maggots

[Calliphoridae] *Chrysomya chloropyga* Maggots

[Calliphoridae] *Chrysomya albiceps* Maggots

[Calliphoridae] *Chrysomya marginalis* Maggots

[Calliphoridae] *Lucilia cuprina* and *L. sericata* Maggots (note that these are not separated, if separate data is able to be produced, create a new folder for that specific species)

[Muscidae] *Synthesiomyia nudiseta* Maggots

Adult Flies

[Calliphoridae] *Chrysomya albiceps*

[Calliphoridae] *Chrysomya marginalis*

[Calliphoridae] *Chrysomya chloropyga*

[Calliphoridae] *Chrysomya megacephala*

[Calliphoridae] *Lucilia sericata* and *L. cuprina* (note that these are not separated, if separate data is able to be produced, create a new folder for that specific species)

[Muscidae] *Synthesiomyia nudiseta*

[Muscidae] (Other)

[Piophilidae]

[Sarcophagidae]

For species that are not represented in these folders, a folder featuring the name of that species should be created wherein all the images featuring said species can be placed. The family should be stated in square brackets. Where possible, the folder name should contain the family, genus and species. The folders should delineate between flies and maggots as is seen in the existing folders. If possible, save the metadata, featuring date taken, author, and camera maker, alongside the photographs in the folders.

Exposure



Photo of *Chrysomya chloropya* with **good lighting**. The veins in the wings are clearly visible and details such as small hairs remain intact.



Photo of *Chrysomya chloropya* with **underexposed lighting**. The veins in the wings are difficult to see and details such as small hairs are not clearly visible.



Photo of *Chrysomya albiceps*. While the lighting is not significantly overexposed, the focus is suboptimal, as details in the wing have become blurry.

Examples of angles



Photo of *Chrysomya chloropyga* focusing on wing



Photo of *Chrysomya chloropyga* focusing on left lateral aspect

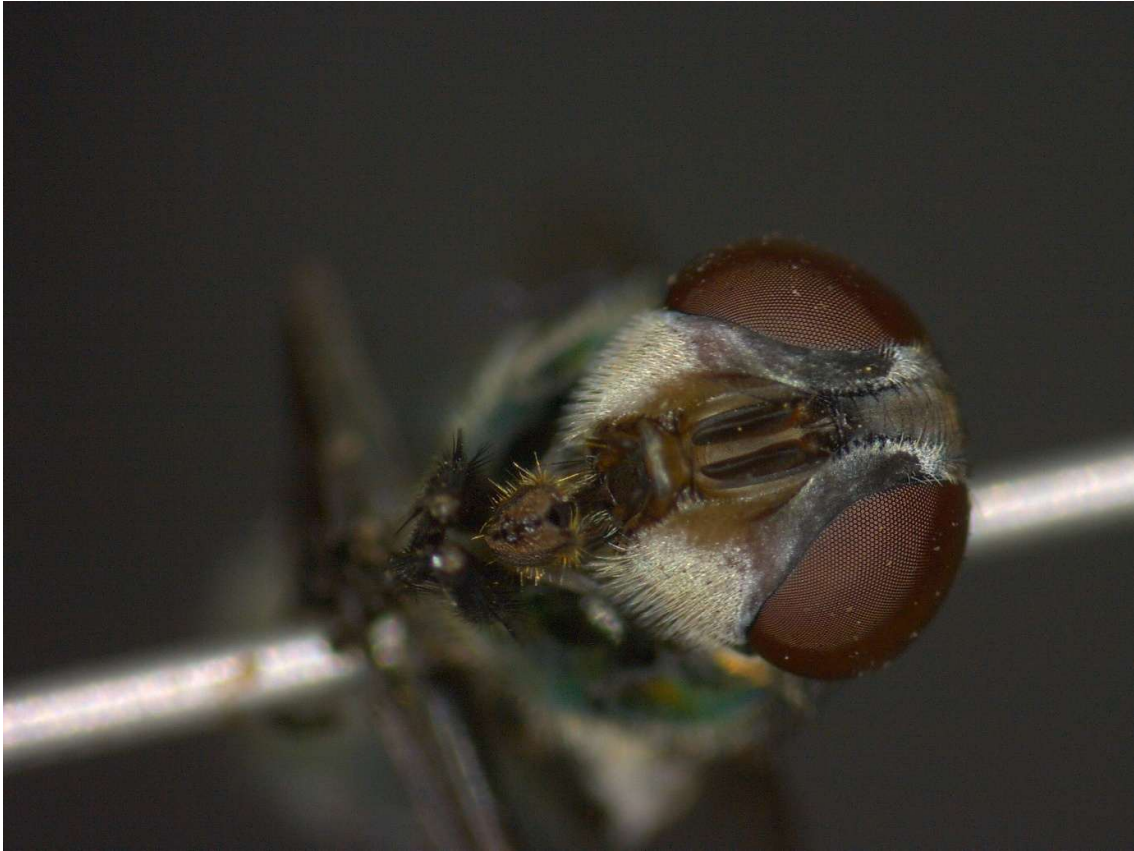


Photo of *Chrysomya chloropyga* focusing on the anterior (face) aspect of the head



Photo of *Chrysomya chloropyga* focusing on dorsal aspect

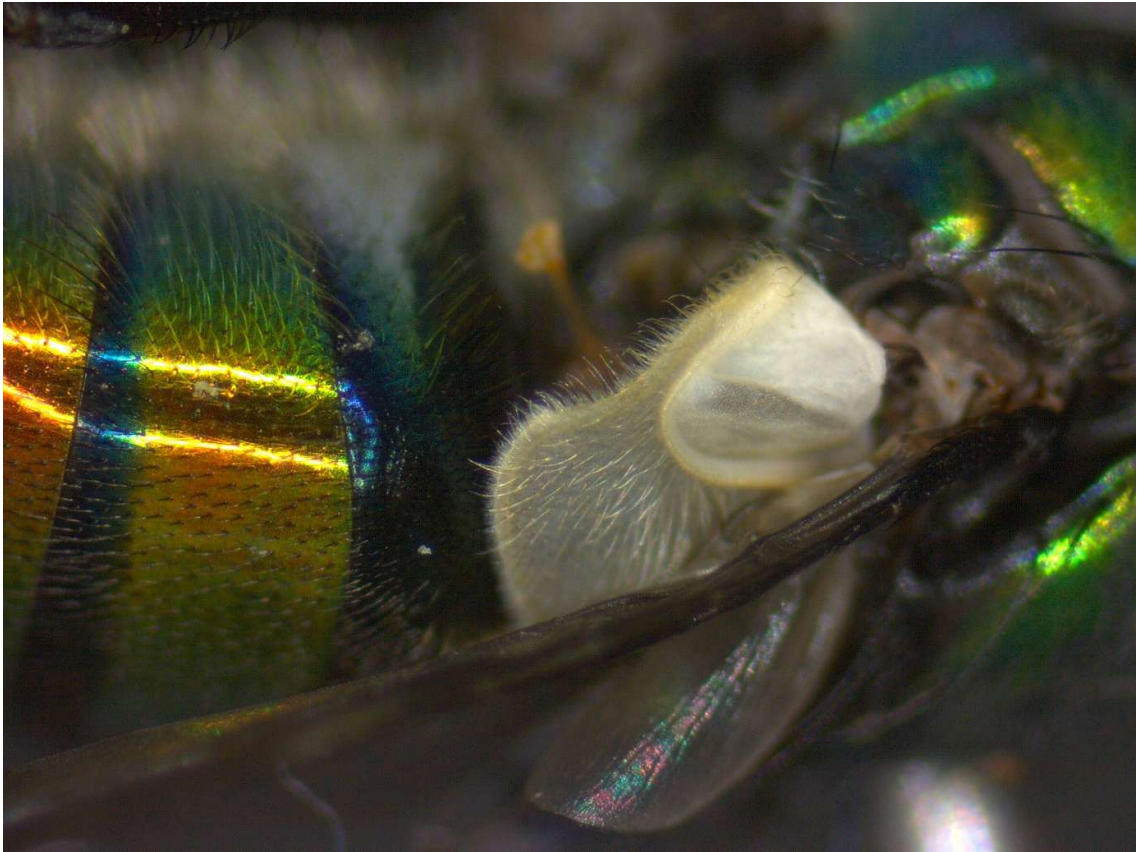


Photo of *Chrysomya albiceps* focusing on calypters